# Topic

"Evaluation of invitro antioxidant, anti inflammatory crude methanolic extract of ***Musa Paradisiaca Inflammatory***"

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# Introduction

Musa paradisiaca, commonly known as plantain, is a tropical plant widely used for both nutritional and medicinal purposes. *Musa paradisiaca*, commonly known as the plantain or cooking banana, is a hybrid banana from the Musaceae family, vital globally as a staple food and used extensively in traditional medicine for its rich nutrients and compounds that support antioxidant, anti-inflammatory, and antidiabetic effects, acting as a giant herbaceous perennial with large leaves forming a false stem and producing nutritious fruit and flowers. It's cultivated worldwide in tropical regions and has significant roles in culinary traditions, from mashing into dishes like *matoke* to industrial uses like fiber production, alongside a vast array of traditional medicinal applications for ailments like ulcers, hypertension, and diarrhea, with scientific studies validating many of these benefits.

# Plant Details:

Scientific Name: Musa paradisiaca

- Common Name: Plantain / Cooking Banana
- Family: Musaceae
- Type: Large, perennial herb (not a true tree)
- Height: 2–7 meters
- Stem: Pseudostem made of leaf sheaths
- Leaves: Large, green, oblong, spirally arranged
- Inflorescence: Purple flower bud with male and female flowers
- Fruit: Long, starchy bananas used in cooking
- Habitat: Tropical and subtropical regions
- Origin: Southeast Asia; widely grown in Bangladesh, India, Africa

# Objective of the Study

This study aims to investigate the pharmacological effects of the crude methanolic extract (CME) from the inflorescence of Musa paradisiaca, a plant known for its traditional medicinal uses. The research focuses on examining its anti-inflamatory activities using in vitro methods. The purpose is to provide scientific evidence supporting its traditional applications and to explore its potential as a natural remedy for conditions related to oxidative stress, inflammation, and cardiovascular disease.

## Study protocol

1. Collection of the plant part.
2. Washing, drying, grinding of the plant part into coarse powder.
3. Extraction of the powdered plant with methanol by the cold extraction method.
4. Determination of the in vitro anti-inflammatory activity of the crude methanolic extract (CME)
5. Determination of the antioxidant activity.

# MATERIALS AND METHODS

## Extraction:

Plant was collected, Washed and sun dried

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Oven dried at 40-45°C

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Grinded into coarse powder

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Methanol was added and Kept 7 days with continuous shaking

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Filtered with cotton and Whatman no 1 filter paper

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Methanol was evaporated At room temperature

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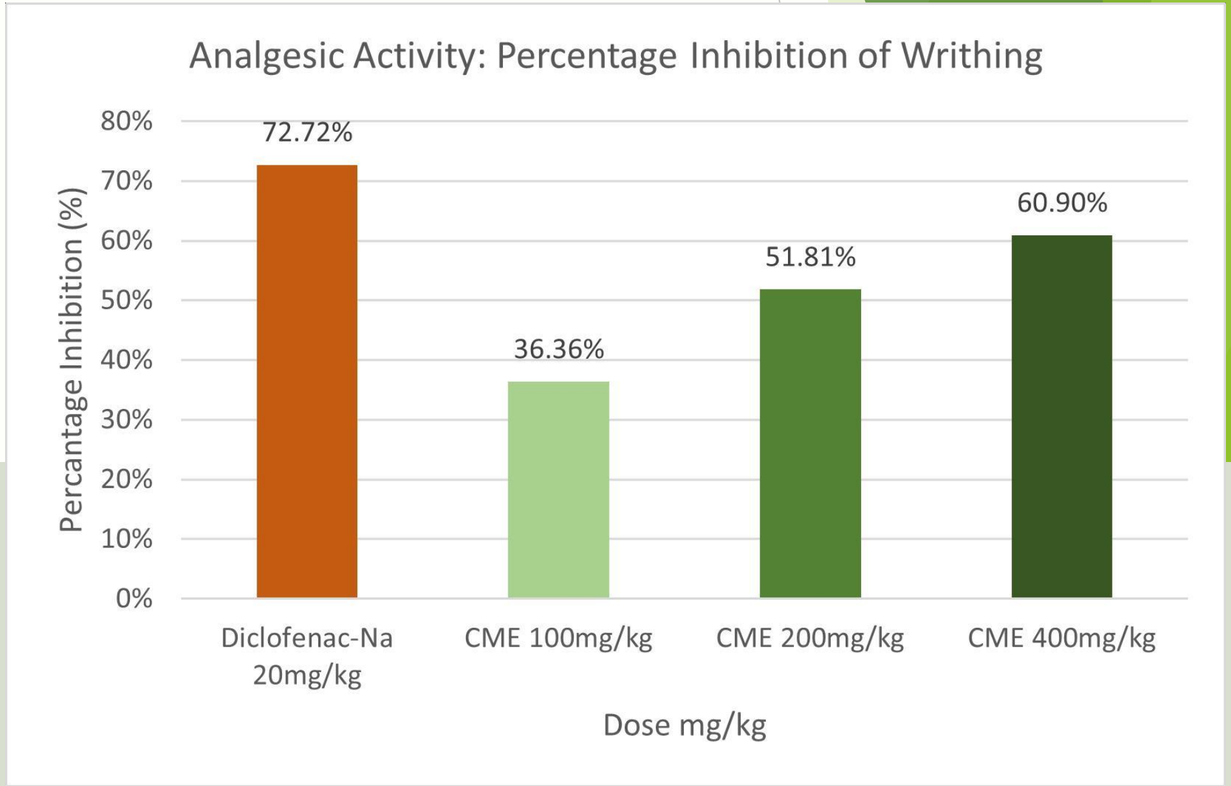
Extract was ready To use

# Result

- **Analgesic activity of methanolic extract of Musa paradisiaca in writhing method**

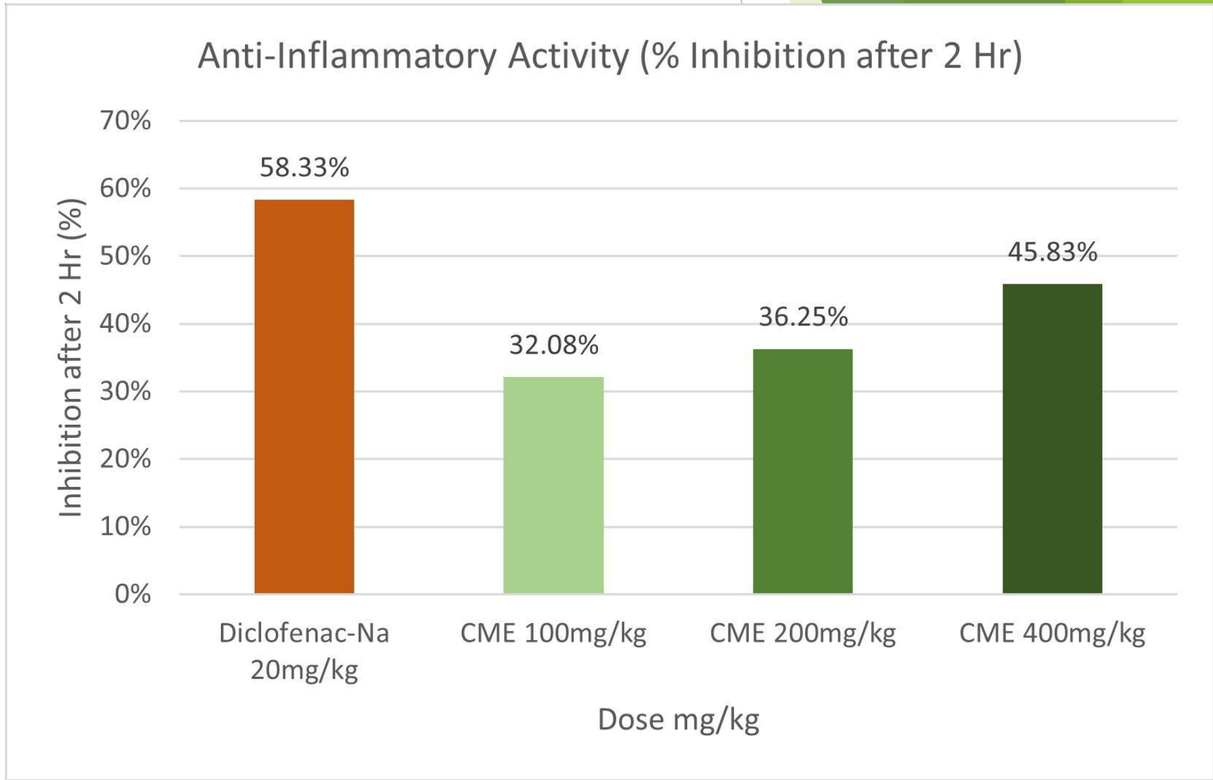
| Sample | Dose (mg/kg) | Writhing Number | % inhibition of writhing number |
|---|---|---|---|
| Control | -------- | 4.0±2.65 | -------- |
| Diclofenac-Na | 20 | 3.0±1.0 | 72.72 |
| CME | 100 | 7.0±1.41 | 36.36 |
| CME | 200 | 3.5±1.15 | 51.81 |
| CME | 400 | 4.3±0.57 | 60.90 |



# Result

- **Anti-Inflammatory Activity of Methanolic Extract of Musa Paradisiaca in Carrageenan-induced Paw Edema Method**

| Sample | Dose (mg/kg) | 0.0 (H) | 0.5 (H) | 1.0 (H) | 2.0 (H) | % inhibition after 2 hours |
|---|---|---|---|---|---|---|
| Control | -------- | 1.1±0.11 | 1.86±0.05 | 2.13±0.05 | 2.4±0.1 | -------- |
| Diclofenac-Na | 20 | 1.0±0.1 | 1.3±0.1 | 1.13±0.05 | 1.0±0.1 | 58.33 |
| CME | 100 | 1.2±0.1 | 1.8±0.1 | 1.6±0.07 | 1.63±0.07 | 32.08 |
| CME | 200 | 1.23±0.15 | 1.9±0.2 | 1.6±0.2 | 1.53±0.2 | 36.25 |
| CME | 400 | 1.33±2.0 | 1.93±0.15 | 1.56±0.2 | 1.3±0.12 | 45.83 |



# Discussion

The crude methanolic extract (CME) of Musa paradisiaca inflorescence demonstrated notable pharmacological properties across all in vitro assays:

1. **Effective Inflammation Reduction:** The study confirms that the methanolic extract of *Musa paradisiaca* (CME) significantly reduces acute inflammation in a dose-dependent manner. While it did not outperform the standard drug Diclofenac-Na, the highest dose (400 mg/kg) achieved a substantial 45.83% inhibition of paw edema .
2. **Targeted Biochemical Pathways:** The extract showed particular effectiveness during the second hour of the experiment. This suggests that its primary mechanism involves interfering with the late-phase inflammatory response, specifically by inhibiting **prostaglandins**, **nitric oxide**, and **cytokines**.
3. **Role of Phytochemicals:** The anti-inflammatory activity is attributed to a synergistic blend of secondary metabolites, including **flavonoids, tannins, and phenolics**. These compounds are known to suppress the cyclooxygenase (COX) and lipoxygenase pathways, which are critical drivers of pain and swelling.
4. **Scientific Support for Traditional Use:** The consistent results and reliable biological response provide a scientific foundation for the traditional use of the banana plant in treating wounds, ulcers, and swelling. It also suggests that the plant's **antioxidant capacity** may play a supportive role in its therapeutic efficacy.

# Conclusion

The methanolic extract of *Musa paradisiaca* shows **significant analgesic and anti- inflammatory activities** in experimental animal models.

- Analgesic activity was **dose-dependent**, with 400 mg/kg performing closest to Diclofenac-Na.
- Anti-inflammatory activity also increased with dose, with the highest effect again at 400 mg/kg.
- These findings suggest that the plant extract contains biologically active compounds capable of reducing pain and inflammation, supporting its traditional medicinal use.

# THANK YOU!